THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
031 PHYSICS

Time: 2½ Hours **November 2025**

Instructions

1. This paper consists of sections **A**, **B** and **C** with a total of **ten** questions.
2. Answer **ALL** questions.
3. Cellular phones and any unauthorized materials are not allowed in the examination room.
4. Whenever required use g= 10m/s^2 or 10N/kg
5. Write your answers neatly.

Question Number	Scores	Examiner's initial
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

SECTION A (15 MARKS)

1. Answer all questions. Choose the most correct answer and write its letter beside the number in your answer booklet.

(i) A branch of Physics which deals which with the study of universe and everything contained in it:

A. Geophysics B. Astronomy C. Electronics D. Mechanics

(ii) When a force is applied to a given material, the material returns to its original shape and size. This behavior is termed as:

A. Plastic deformation
B. Elastic Deformation
C. Irreversible deformation
D. Viscous deformation

(iii) Which instrument would give the most accurate measurement of the thickness of a coin?

A. Metre rule B. Vernier caliper C. Tape measure D. Ruler E. Protractor

(iv) Which expression has the same units as the SI unit of Pressure:

A. $\frac{force}{length \times speed}$ B. $\frac{fore}{length \times time}$ C. $\frac{mass}{length \times (time)^2}$ D. $\frac{mass \times (time)^2}{length}$

(v) The displacement of an object for a round a trip between two locations

A. is always greater than zero
B. is always greater than zero
C. zero
D. Can have any value

(vi) Work is said to be done when:

A. Force acts upon a body but the body does not move
B. Force acts upon a body and it moves in the direction of force applied
C. Force acts upon a body but the body does not move in the direction of the force
D. Force acts upon the body and the body moves in opposite to the direction of force applied

(vii) A liquid has a volume of $100cm^3$ and mass of 85g. The density of water is $1.0\frac{g}{cm^3}$. How does the density of the liquid compare with the density of water?

A. Its density is greater than that of water
B. Its density is lower than of water
C. Its density is the same as that of water
D. It is difficult to determine with only this data

(viii) The sea water is denser than freshwater due to

A. Mixing of sand B. Stagnation C. Mixing salts D. Evaporation

(ix) If the radius of a capillary tube is reduced while keeping all other factors constant, how will it affect the capillary rise height?

A. The capillary rise will increase.
B. The capillary rise will decrease.
C. The capillary rise will remain the same.
D. The capillary rise will become unpredictable.

(x) The relation of physics with other sciences is best described as:

A. Physics is an isolated science
B. Physics is used only in chemistry
C. Physics connects many scientific principles
D. Physics is only related to biology

Answers

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)

2. Match the items in list A with the correct response in list B. (5marks)

List A	List B
(i) Surface tension (ii) Capillarity (iii)Cohesive forces (iv)Adhesive force (v) Elasticity	A. A force exerted by a surface that acts perpendicular to the surface when an object is in contact with it. B. A force that resists the relative motion of objects through a fluid (liquid or gas). C. Water droplets form spherical shapes due to this property. D. Water rises in a thin glass tube, or in the tiny tubes of plant roots and stems to this property. E. A pulling force transmitted through a string, cable, or rope. F. The attraction between water molecules that keeps them together. G. A rubber band returns to its original length after being stretched. H. The attraction between water molecules and the glass surface of a tube.

Answers

(i)	(ii)	(iii)	(iv)	(v)

SECTION B (70MARKS)

3. You are provided with different measuring instruments such as a ruler, stopwatch, thermometer, measuring cylinder, and beam balance. Using your knowledge of measurement in physics, identify five physical quantities that can be measured with such instruments and state their corresponding SI units

Physical quantity	Instrument used	SI unit
(i)		
(ii)		
(iii)		
(iv)		
(v)		

4. (a) Explain three conditions for a body to float:

(i) ……

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(ii) ……

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(iii)……

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(b) Calculate the density of a body if a floating body is 95% submerged in water. The density of water is $1000\,\frac{kg}{m^3}$

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(c)In an experiment to determine the relative density of the liquid, a solid Q weighed as follows: weight in air W_0= 8.6 N, weight in water W_1= 6.0 N, weight in Liquid W_2= 5.4 N. Determine the upthrust of the liquid.

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5. (a) State the kinetic theory of matter.

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(b) List three factors that affect the surface tension of given liquid

(i) ……

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(ii) ……

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(iii)……

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(c) Give five applications of capillarity in daily life

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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(v) ……

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6. (a) A block of wood weighs 3N and measures 5 cm × 10 cm × 4 cm. If it is placed on a table with its largest base area or smallest area face in contact with the table, it would exert different pressure on the table. Which surface would exert the largest pressure on the table? Explain.

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(b) State and explain the principle by which a hydraulic press operates, and mention one of its applications

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(c) Explain two factors which affect pressure in liquids

(i) ……

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(ii) ……

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7. (a) Explain the difference between kinetic energy and potential energy.

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(b) What is the kinetic energy of a 12g bullet travelling at a speed of 320m/s.

(c) Mention five forms of energy you have learned about

(i) ……

(ii) ……

(iii)……

(iv)……

(v) …….

8. (a) Give the difference between:

(i) Distance and displacement

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(ii) Speed and velocity

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(b) A car starts from rest and acerates uniformly at the of $4\frac{m}{s^2}$ for 5s. It maintains the constant velocity for 20s the brakes are then applied and the car decelerates for 3s; find:

(i) The maximum velocity reached

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(ii) The total distance covered

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9. (a) A train travelling at $36km/h$ stops when its brakes are applied. The train suffers a deceleration of $2m/s^2$

(i) How long does it take to come to rest

(ii) What is the final velocity

(b) A body moved vertically upward to a maximum height of 20m. Calculate the initial velocity

SECTION C (15MARKS)

10. (a) During a science lesson, a student asked why astronauts feel lighter on the Moon even though their bodies remain the same. The teacher responded by discussing the concepts of mass and weight. Based on this scenario:

(i) What misunderstanding about mass and weight do you think the student had?

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(ii) Explain the differences between mass and weight using the example of the Earth and the Moon

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(b) Describe five practical applications of density or relative density in everyday life.

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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(v) ……

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(c)In an experiment to determine the density of liquid X, a Form One student recorded the following results: Mass of beaker = 600 g and Mass of beaker + liquid (25 cm^3) = 700 g. What was the student's calculated density for liquid X?

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